

**32** Aortic stenosis is a heart valve disorder in which the aortic semilunar valve opening is narrow.

Which effect could aortic stenosis have on heart structure and function?

- A** The cardiac muscle of the left ventricle wall is thinned by blood leaking out of the left ventricle during ventricular diastole.
- B** There is less cardiac muscle in the left ventricle and reduced diastolic blood pressure, caused by the smaller blood volume entering the left atrium.
- C** The tendons of the heart valves are weakened by blood being forced back through the left atrioventricular valve into the left atrium.
- D** The wall of the left ventricle thickens, leading to an enlarged heart and inability to relax and fill completely during diastole.